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CHAYHANA: Starpplatformu mobilā CRM un rezervāciju automatizācijas sistēma premium tējnīcām ar vairāku lomu piekļuvi un reāllaika analītiku

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CHAYHANA: Cross-Platform Mobile CRM and Reservation Automation System for Premium Teahouses with Multi-Role Access and Real-Time Analytics

BACHELOR'S THESIS

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Anotācija

Šis bakalaura darbs pievēršas rezervāciju pārvaldības un klientu attiecību procesu modernizācijai premium tējnīcu iestādēs, izstrādājot starpplatformu mobilo lietojumprogrammu sistēmu. Pētījums koncentrējas uz CHAYHANA — visaptverošas CRM un rezervāciju automatizācijas platformas projektēšanu, ieviešanu un novērtēšanu, kas veidota, izmantojot Flutter ietvaru ar Firebase Cloud Firestore aizmugursistēmu.

Bakalaura darbs sastāv no 5 nodaļām. 1. nodaļā aplūkoti CRM sistēmu, mobilo lietojumprogrammu izstrādes ietvaru un rezervāciju automatizācijas tehnoloģiju teorētiskie pamati viesmīlības nozarē. 2. nodaļā sniegta esošo rezervāciju un CRM risinājumu analīze, identificējot to ierobežojumus. 3. nodaļā aprakstīta sistēmas arhitektūras projektēšana, datubāzes shēma un vairāku lomu piekļuves kontroles modelis. 4. nodaļā detalizēta starpplatformu mobilās lietojumprogrammas ieviešana ar pamata moduļiem. 5. nodaļā sniegti testēšanas rezultāti un sistēmas efektivitātes novērtējums.

Pētījuma metodoloģiskais pamats ietver zinātniskās literatūras analīzi, ieinteresēto pušu intervijas un nozares pārskatus. Izmantotās metodes: sistemātisks literatūras apskats, salīdzinošā analīze, sistēmas izstrāde, lietojamības testēšana, veiktspējas salīdzinošā novērtēšana un eksperimentālā novērtēšana.

Galvenie rezultāti parāda, ka izstrādātā CHAYHANA sistēma efektīvi risina identificētās operacionālās problēmas premium tējnīcu pārvaldībā, nodrošinot vairāku lomu piekļuves kontroli, reāllaika analītiku un netraucētu rezervāciju automatizāciju caur intuitīvu starpplatformu mobilo saskarni.

Bakalaura darbs izstrādāts **XX** lapaspusēs un ietver **YY** tabulas, **ZZ** attēlus, **xy** algoritmus, **yz** programmu kodu izdrukas (listings), **xz** pielikumus, kā arī literatūras sarakstu ar **zx** avotiem.

Atslēgvārdi: starpplatformu mobilā lietojumprogramma, CRM sistēma, rezervāciju automatizācija, Flutter ietvars, Firebase, tējnīcu pārvaldība, reāllaika analītika.

Abstract

This Bachelor Paper addresses the challenge of modernizing reservation management and customer relationship processes in premium teahouse establishments through the development of a cross-platform mobile application system. The study focuses on the design, implementation, and evaluation of CHAYHANA — a comprehensive CRM and reservation automation platform built using Flutter framework with Firebase Cloud Firestore backend.

The Bachelor Paper consists of 5 chapters. Chapter 1 reviews the theoretical background of CRM systems, mobile application development frameworks, and reservation automation technologies in the hospitality industry. Chapter 2 presents the analysis of existing reservation and CRM solutions, identifying their limitations and gaps relevant to premium teahouse operations. Chapter 3 describes the system architecture design, database schema, and multi-role access control model. Chapter 4 details the implementation of the cross-platform mobile application with core modules including reservation management, real-time notifications, and analytics dashboard. Chapter 5 presents the testing results, performance benchmarking, and evaluation of the system's effectiveness.

The methodological basis of the study includes the analysis of scientific literature, stakeholder interviews, and industry reports. The applied methods are: systematic literature review, comparative analysis, system development and engineering, usability testing using System Usability Scale (SUS), performance benchmarking, and experimental evaluation.

The main results demonstrate that the developed CHAYHANA system effectively addresses the identified operational challenges in premium teahouse management, providing multi-role access control, real-time analytics, and seamless booking automation through an intuitive cross-platform mobile interface.

The Bachelor Paper is written in **XX** pages and contains **YY** tables, **ZZ** figures, **xy** algorithms, **yz** listings, **xz** annexes, and a list of references with **zx** sources.

Keywords: cross-platform mobile application, CRM system, reservation automation, Flutter framework, Firebase, teahouse management, real-time analytics.

Key words / Atslēgvārdi

English	Latviešu
Cross-platform mobile application	Starpplatformu mobilā lietojumprogramma
CRM system	CRM sistēma
Reservation automation	Rezervāciju automatizācija
Flutter framework	Flutter ietvars
Firebase Cloud Firestore	Firebase Cloud Firestore
Multi-role access control	Vairāku lomu piekļuves kontrole
Real-time analytics	Reāllaika analītika

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Introduction

Topicality of the problem: The premium teahouse industry, particularly in Central Asian markets, is experiencing rapid growth and modernization. However, most establishments still rely on manual processes for reservation management, customer relationship management, and business analytics. This creates significant operational challenges including booking conflicts, missed personalization opportunities, and insufficient data for strategic decision-making. The absence of integrated digital solutions specifically designed for premium teahouse establishments represents both a market gap and a research opportunity. With the proliferation of mobile technology and cloud-based services, there is an urgent need for a comprehensive cross-platform solution that addresses these challenges while accommodating the unique operational requirements of premium teahouses.

Aim of the study: To design, develop, and deploy a comprehensive cross-platform mobile Customer Relationship Management (CRM) and reservation automation system specifically tailored for premium teahouse establishments. The system will streamline operations through multi-role access control (customer, staff, administrator), real-time analytics, and seamless booking management, ultimately enhancing customer experience, operational efficiency, and business intelligence for teahouse owners.

Object of the study: The reservation and customer relationship management process in premium teahouse establishments—encompassing the entire customer journey from discovery and booking to visit confirmation and post-service feedback, as well as the administrative processes of resource management, analytics generation, and business intelligence utilization.

Subject of the study: The development, implementation, and evaluation of a cross-platform mobile application system utilizing Flutter framework, Firebase Cloud Firestore, Firebase Authentication, and push notification services, specifically designed for premium teahouse reservation automation and CRM with multi-role access control and real-time analytics capabilities.

Research problem: Premium teahouses currently face significant operational challenges including:

1. Manual reservation management leading to booking conflicts and lost revenue.

2. Lack of integrated customer relationship management causing missed personalization opportunities.
3. Insufficient real-time business analytics preventing data-driven decision making.
4. Absence of role-based access control creating security and workflow inefficiencies.
5. Poor cross-platform compatibility limiting customer accessibility and staff mobility.

These issues result in decreased customer satisfaction, operational inefficiencies, and reduced competitive advantage in the premium hospitality sector.

Tasks of the study:

1. To conduct a systematic literature review and comparative analysis of existing CRM and reservation management systems in the hospitality industry, identifying their limitations and gaps relevant to premium teahouse operations.
2. To analyze and define the functional and non-functional requirements for a cross-platform mobile CRM and reservation automation system based on stakeholder needs (customers, staff, administrators).
3. To design the system architecture, database schema, and multi-role access control model using Flutter framework, Firebase Cloud Firestore, and Firebase Authentication.
4. To develop and implement the cross-platform mobile application with core modules: reservation management, customer relationship management, real-time notifications, and analytics dashboard.
5. To implement and validate the real-time analytics engine, enabling data-driven decision-making through automated report generation and key performance indicator (KPI) tracking.
6. To perform comprehensive testing including usability testing, performance benchmarking, and security assessment of the developed system.
7. To evaluate the system's effectiveness by comparing operational metrics (booking conflicts, processing time, customer satisfaction) before and after system deployment against the stated hypotheses.

Hypothesis: A well-designed cross-platform mobile CRM and reservation automation system with multi-role access and real-time analytics will:

1. Reduce booking conflicts and manual errors by at least 80% compared to traditional manual management.
2. Increase operational efficiency through automated workflows and instant notifications.
3. Improve customer satisfaction scores through seamless booking experience and personalized service.
4. Provide actionable business insights through real-time analytics, enabling data-driven decision making.
5. Demonstrate superior performance and usability compared to existing generic reservation systems.
6. Achieve high adoption rates among both customers and teahouse staff due to intuitive cross-platform interface.

Research methods:

Qualitative Research (Inductive):

- **Literature Review:** Systematic review of academic publications and industry reports on CRM systems, reservation automation, and mobile application development in the hospitality sector.
- **Stakeholder Analysis:** Interviews and requirements gathering from teahouse owners, staff, and customers to define functional specifications.

Quantitative Research (Deductive/Experimental):

- **System Development/Engineering:** Designing and implementing the cross-platform mobile application using Flutter, Firebase Cloud Firestore, and Firebase Authentication.
- **Usability Testing:** Conducting user experience evaluations using System Usability Scale (SUS) methodology with target user groups.

- **Performance Benchmarking:** Measuring system response times, concurrent user handling, and real-time data synchronization reliability.
- **Comparative Analysis:** Evaluating the developed system against existing reservation solutions based on functionality, performance, and user satisfaction metrics.

1. Literature Review and Theoretical Background

1.1. Core Concepts and Definitions

1.1.1. Customer Relationship Management (CRM)

Customer Relationship Management (CRM) is a strategic approach that organizations use to manage interactions with current and potential customers. According to [2], CRM encompasses the practices, strategies, and technologies that companies employ to track and analyze customer interactions and data throughout the customer lifecycle. The primary goal of CRM is to improve customer service relationships, assist in customer retention, and drive sales growth.

In the context of the hospitality industry, CRM systems play a critical role in managing guest preferences, booking histories, and personalized service delivery. Modern CRM solutions integrate multiple communication channels, automate routine tasks, and provide analytical capabilities for data-driven decision making [7].

1.1.2. Reservation Management Systems

Reservation management systems are software applications designed to handle the booking and scheduling of services or resources. In the hospitality sector, these systems manage table reservations, room bookings, and event scheduling. Traditional reservation systems were primarily phone-based or walk-in oriented, but digital transformation has shifted the paradigm toward online and mobile-based solutions [3].

Key features of modern reservation systems include real-time availability checking, automated confirmation and reminder notifications, capacity management, and integration with other business systems such as point-of-sale (POS) and CRM platforms.

1.1.3. Cross-Platform Mobile Development

Cross-platform mobile development refers to the practice of building mobile applications that can run on multiple operating systems (primarily iOS and Android) from a single codebase. This approach significantly reduces development time and costs compared to native development, while maintaining near-native performance and user experience [1].

Flutter, developed by Google, is a leading cross-platform framework that uses the Dart programming language. It provides a rich set of pre-built widgets, hot reload capability

for rapid development, and compiles to native ARM code for optimal performance [9].

1.1.4. Firebase and Cloud-Based Backend Services

Firebase is a comprehensive platform developed by Google for building mobile and web applications. It provides a suite of cloud-based services including Cloud Firestore (a NoSQL document database), Authentication, Cloud Storage, Cloud Functions, and Cloud Messaging [5].

Cloud Firestore offers real-time data synchronization, offline support, and automatic scaling, making it particularly suitable for applications requiring real-time updates and multi-user collaboration. Firebase Authentication provides secure user identity management with support for multiple authentication providers.

1.2. Related Work: Analysis of Existing Solutions

1.2.1. General Hospitality CRM and Reservation Systems

Several reservation and CRM platforms currently serve the hospitality industry. Systems such as OpenTable, Resy, and Yelp Reservations dominate the restaurant reservation market in Western countries. These platforms offer features including online booking, wait-list management, guest profiling, and basic analytics [4].

However, these solutions share common limitations:

- **Geographic focus.** Most established platforms are designed for Western markets and lack localization for Central Asian markets.
- **Generic approach.** They are designed for general restaurants and do not address the unique operational requirements of premium teahouse establishments.
- **Limited role management.** Many solutions offer basic user roles without comprehensive multi-tier access control needed for complex organizational structures.
- **High costs.** Subscription-based pricing models make them inaccessible for small and medium-sized establishments in developing markets.

1.2.2. Mobile-First Reservation Applications

The trend toward mobile-first solutions has led to the development of numerous reservation applications. Studies by [8] demonstrate that mobile booking applications significantly increase reservation rates and customer satisfaction compared to web-only solutions. Key

advantages include push notification capabilities, location-based services, and seamless payment integration.

1.2.3. Comparative Analysis of Existing Solutions

Table 1.1. Comparison of existing reservation and CRM systems

Feature	OpenTable	Resy	Generic CRM	CHAYHANA
Cross-platform mobile	Partial	Yes	Varies	Yes
Real-time analytics	Basic	Basic	Advanced	Advanced
Multi-role access	Limited	Limited	Yes	Yes
Teahouse-specific	No	No	No	Yes
Offline support	No	No	Varies	Yes
Central Asian market	No	No	No	Yes

Source: Author's compilation based on platform analysis, 2026.

1.3. Theoretical Framework

The theoretical foundation of this study draws upon several established frameworks:

Technology Acceptance Model (TAM). Originally proposed by Davis (1989), TAM suggests that perceived usefulness and perceived ease of use are fundamental determinants of user acceptance of information technology. This framework guides the design decisions for the CHAYHANA system's user interface and feature set.

Service Quality Model (SERVQUAL). The SERVQUAL model provides a framework for measuring service quality across five dimensions: tangibles, reliability, responsiveness, assurance, and empathy. This model informs the evaluation criteria for assessing the system's impact on teahouse service quality.

Agile Software Development Methodology. The system development follows Agile principles with iterative development cycles, continuous user feedback integration, and incremental feature delivery. This approach ensures that the final product closely aligns with stakeholder requirements.

1.4. Research Motivation and Direction

Problem significance. The premium teahouse sector represents a significant and growing segment of the hospitality industry in Central Asian markets. Despite this growth, the sector lacks dedicated digital solutions that address its unique operational requirements, resulting in inefficiencies that directly impact revenue and customer satisfaction.

Identified gap. The analysis of existing solutions reveals that no current platform provides a comprehensive, cross-platform mobile CRM and reservation system specifically designed for premium teahouse establishments with multi-role access control and real-time analytics capabilities adapted for Central Asian markets.

Direction and scope. This thesis addresses the identified gap by designing, developing, and evaluating the CHAYHANA system — a purpose-built cross-platform mobile application that combines CRM functionality, reservation automation, and real-time analytics in a single platform optimized for premium teahouse operations.

Success criteria. The system will be evaluated based on: (1) reduction in booking conflicts, (2) improvement in operational efficiency metrics, (3) user satisfaction scores using SUS methodology, and (4) system performance benchmarks including response time and concurrent user handling.

1.5. Chapter Summary

This chapter established the conceptual foundation by defining key terms including CRM, reservation management systems, cross-platform mobile development, and cloud-based backend services. A comparative analysis of existing solutions revealed significant gaps in the market for teahouse-specific digital management tools. The theoretical framework combining TAM, SERVQUAL, and Agile methodology provides the basis for the system's design, development, and evaluation.

Conclusions (for Chapter 1). The literature review confirms that while numerous CRM and reservation systems exist for the hospitality industry, none adequately addresses the specific needs of premium teahouse establishments in Central Asian markets. This gap motivates the development of the CHAYHANA system, which will be analyzed in detail in Chapter 2.

2. Requirements Analysis and System Specifications

2.1. Stakeholder Analysis

The CHAYHANA system serves three primary stakeholder groups, each with distinct needs and expectations:

2.1.1. Customers (End Users)

Customers represent the largest user base of the system. Their primary needs include:

- Discovering premium teahouses based on location, ratings, and available amenities.
- Making reservations quickly and conveniently through a mobile application.
- Viewing booking history and managing upcoming reservations.
- Providing feedback and reviews after their visits.
- Receiving real-time notifications about booking confirmations, reminders, and special offers.

2.1.2. Teahouse Owners and Administrators

Owners and administrators require comprehensive management capabilities:

- Managing teahouse profiles, menus, and available rooms or areas.
- Viewing and managing incoming reservations with real-time updates.
- Accessing analytics dashboards with key performance indicators (KPIs).
- Managing staff accounts and access permissions.
- Generating reports on revenue, occupancy rates, and customer satisfaction.

2.1.3. Staff Members

Staff members need streamlined operational tools:

- Viewing daily reservation schedules and guest details.
- Updating reservation statuses (confirmed, in-progress, completed).

- Processing orders and managing table/room assignments.
- Communicating with customers regarding their bookings.

2.2. Functional Requirements

Based on stakeholder analysis and industry best practices, the following functional requirements have been identified:

Table 2.1. Functional requirements specification

ID	Category	Description
FR-01	Authentication	User registration and login with email/password and social media options
FR-02	Authentication	Multi-role access control (Customer, Staff, Owner, Super Admin)
FR-03	Booking	Real-time reservation creation with date, time, and guest count selection
FR-04	Booking	Automated booking confirmation and reminder notifications
FR-05	Booking	Booking history and management for customers
FR-06	CRM	Customer profile management with visit history
FR-07	CRM	Review and rating system for teahouses
FR-08	Analytics	Real-time dashboard with KPI visualization
FR-09	Analytics	Revenue and occupancy reports generation
FR-10	Management	Teahouse profile and menu management

Source: Author's compilation based on stakeholder analysis, 2026.

2.3. Non-Functional Requirements

2.4. Use Case Analysis

The following use case diagram illustrates the primary interactions between actors and the system:

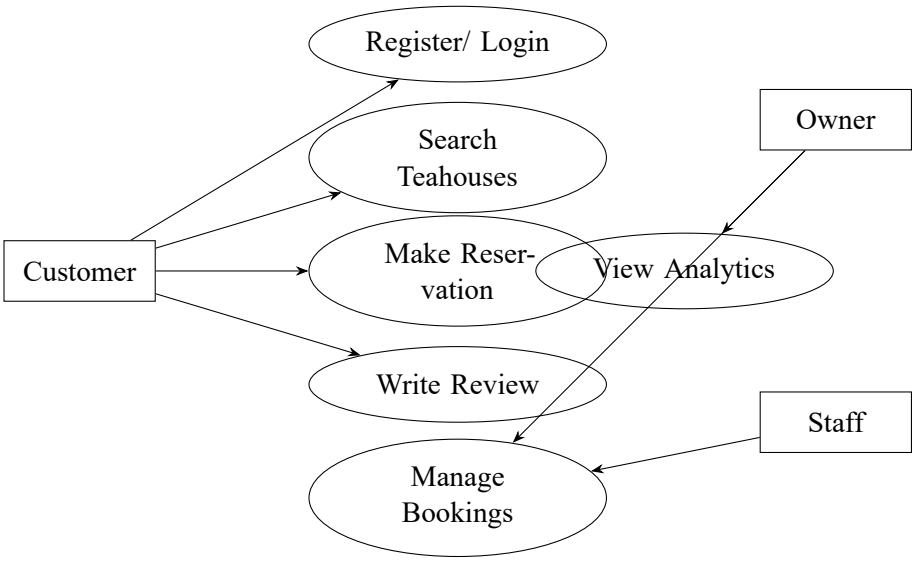
2.5. Technology Stack Selection

Based on the requirements analysis, the following technology stack was selected:

Table 2.2. Non-functional requirements specification

ID	Category	Description
NFR-01	Performance	Application response time under 2 seconds for standard operations
NFR-02	Performance	Support for at least 100 concurrent users
NFR-03	Reliability	99.5% system uptime availability
NFR-04	Security	End-to-end encryption for sensitive data
NFR-05	Security	Role-based access control enforcement
NFR-06	Usability	Intuitive UI with SUS score above 70
NFR-07	Compatibility	Support for Android 8.0+ and iOS 13+
NFR-08	Scalability	Cloud-based architecture supporting horizontal scaling

Source: Author's compilation, 2026.



Source: Author-created diagram.

Figure 2.1. Use case diagram of the CHAYHANA system

- **Frontend Framework:** Flutter 3.x — providing cross-platform compatibility, rich widget library, and native performance compilation.
- **Programming Language:** Dart — object-oriented, optimized for UI development, strong typing support.
- **Backend/Database:** Firebase Cloud Firestore — real-time NoSQL database with offline support and automatic synchronization.
- **Authentication:** Firebase Authentication — secure identity management with multiple provider support.

- **Push Notifications:** Firebase Cloud Messaging (FCM) — cross-platform notification delivery.
- **State Management:** Provider/Riverpod — reactive state management for Flutter applications.
- **Version Control:** Git with GitHub — collaborative development and code versioning.

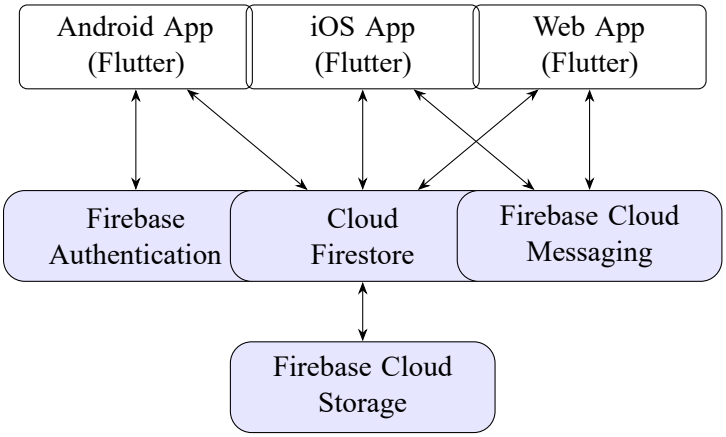
2.6. Chapter Summary

This chapter presented a comprehensive requirements analysis for the CHAYHANA system. Through stakeholder analysis, functional and non-functional requirements were identified and documented. Use case analysis provided a clear visualization of system interactions, and the technology stack was selected based on the requirements. These specifications form the foundation for the system architecture design presented in Chapter 3.

3. System Architecture and Design

3.1. High-Level System Architecture

The CHAYHANA system follows a client-server architecture pattern with a cloud-based backend. The architecture is designed to ensure scalability, reliability, and real-time data synchronization across all connected clients.



Source: Author-created diagram.

Figure 3.1. High-level system architecture of CHAYHANA

The architecture comprises three main layers:

1. **Presentation Layer** — Flutter-based cross-platform mobile and web applications providing the user interface for all user roles.

2. **Business Logic Layer** — Application logic implemented within the Flutter application using state management patterns (Provider/Riverpod).

3. **Data Layer** — Firebase Cloud Firestore providing real-time NoSQL database services with automatic synchronization and offline capabilities.

3.2. Database Schema Design

The Cloud Firestore database is organized using a document-collection model. The main collections and their relationships are defined as follows:

Table 3.1. Primary database collections and their fields

Collection	Key Fields	Description
`users`	uid, email, role, name, phone	User accounts with role-based access
`choyxonas`	id, name, address, rating, ownerId	Teahouse establishment profiles
`bookings`	id, userId, choyxonaId, date, status	Reservation records
`reviews`	id, userId, choyxonaId, rating, text	Customer reviews and ratings
`menus`	id, choyxonaId, items, prices	Menu items for each teahouse
`analytics`	choyxonaId, date, metrics	Aggregated analytics data

Source: Author's database design, 2026.

3.2.1. Collections Structure

3.2.2. Data Relationships

The NoSQL document model uses reference fields to establish relationships between collections. Each booking document references both a user document and a choyxona document through their respective IDs. This denormalized approach optimizes read performance while maintaining data consistency through Cloud Firestore security rules and transaction operations.

3.3. Multi-Role Access Control Model

The system implements a hierarchical role-based access control (RBAC) model with four distinct roles:

Table 3.2. Role-based access control matrix

Permission	Customer	Staff	Owner	Super Admin
View teahouses	✓	✓	✓	✓
Make booking	✓	–	–	✓
Write review	✓	–	–	✓
Manage bookings	–	✓	✓	✓
View analytics	–	–	✓	✓
Manage menu	–	✓	✓	✓
Manage staff	–	–	✓	✓
Manage all teahouses	–	–	–	✓
Manage users	–	–	–	✓

Source: Author's design, 2026.

3.4. Application Navigation Architecture

The application uses a screen-based navigation architecture with role-dependent routing. Upon authentication, users are directed to role-specific home screens:

- **Customer** → Home screen with teahouse discovery, search, and booking functionality.
- **Owner** → Owner dashboard with analytics, reservation management, and business settings.
- **Staff** → Staff panel with daily bookings and order management.
- **Super Admin** → Administrative panel with full system management capabilities.

3.5. Real-Time Data Synchronization

One of the key architectural decisions is the use of Firestore's real-time listeners for data synchronization. This enables:

- Instant reflection of booking status changes across all connected clients.
- Real-time analytics updates on the owner dashboard.
- Live notification delivery for booking confirmations and reminders.
- Offline data access with automatic synchronization upon reconnection.

The synchronization mechanism uses Firestore's `snapshots()` stream, which provides a continuous data feed that automatically updates the UI when underlying data changes.

3.6. Chapter Summary

This chapter presented the system architecture design for CHAYHANA, including the three-layer architecture, database schema with six primary collections, multi-role access control model, navigation architecture, and real-time synchronization mechanism. The architecture decisions directly address the requirements identified in Chapter 2 and provide the blueprint for the implementation described in Chapter 4.

4. System Implementation

4.1. Development Environment Setup

The CHAYHANA system was developed using the following development environment:

Table 4.1. Development environment specifications

Component	Specification
IDE	Visual Studio Code with Flutter/Dart extensions
Flutter SDK	Version 3.x (stable channel)
Dart SDK	Version 3.x
Target Platforms	Android 8.0+, iOS 13+, Web
Backend	Firebase (Blaze Plan)
Version Control	Git with GitHub
Testing	Android Emulator, Physical devices

Source: Author's development setup, 2026.

4.2. Project Structure

The Flutter project follows a feature-based folder structure to ensure maintainability and scalability:

Listing 4.1. CHAYHANA project directory structure

```
lib/
  main.dart                   # Application entry point
  models/                     # Data models
    user_model.dart
    choyxona_model.dart
    booking_model.dart
    review_model.dart
  screens/                    # UI screens
    auth/                     # Authentication screens
    home/                     # Customer home screen
    booking/                  # Booking screens
    choyxona_details/         # Teahouse details
```

```
    owner/                    # Owner dashboard
    reports/                  # Analytics reports
  services/                   # Business logic services
    auth_service.dart
    firestore_service.dart
    notification_service.dart
  widgets/                    # Reusable UI components
    receipt_dialog.dart
    custom_button.dart
  utils/                      # Utility functions
```

4.3. Authentication Module Implementation

The authentication module implements Firebase Authentication with email/password login and role-based routing:

Listing 4.2. Authentication service implementation (Dart)

```
class AuthService {
  final FirebaseAuth _auth = FirebaseAuth.instance;
  final FirebaseFirestore _firestore =
      FirebaseFirestore.instance;

  Future<UserModel?> signIn(
    String email, String password
  ) async {
    final credential =
        await _auth.signInWithEmailAndPassword(
      email: email, password: password,
    );
    final doc = await _firestore
        .collection('users')
        .doc(credential.user!.uid)
        .get();
    return UserModel.fromFirestore(doc);
  }
```

```

  void routeByRole(UserModel user,
      BuildContext context) {
    switch (user.role) {
      case 'customer':
        Navigator.pushReplacement(context,
          HomeScreen());
        break;
      case 'owner':
        Navigator.pushReplacement(context,
          OwnerDashboard());
        break;
      case 'admin':
        Navigator.pushReplacement(context,
          AdminPanel());
        break;
    }
  }
}
```

4.4. Reservation Management Module

The reservation module handles the complete booking lifecycle from creation to completion. Key implementation features include:

- **Date and time validation** — preventing double bookings through real-time availability checking against existing reservations in Firestore.
- **Status management** — bookings transition through states: `pending` → `confirmed` → `in_progress` → `completed` or `cancelled`.
- **Push notifications** — automated notifications sent via Firebase Cloud Messaging (FCM) for booking confirmations, reminders (24 hours before), and status updates.

Listing 4.3. Booking creation with conflict detection (Dart)

```
Future<bool> createBooking(Booking booking) async {
```

```
  // Check for conflicts
  final conflicts = await _firestore
      .collection('bookings')
      .where('choyxonaId',
        isEqualTo: booking.choyxonaId)
      .where('date', isEqualTo: booking.date)
      .where('status',
        whereIn: ['confirmed', 'pending'])
      .get();

  if (conflicts.docs.length >= maxCapacity) {
    throw BookingConflictException(
      'No available slots');
  }

  await _firestore
      .collection('bookings')
      .add(booking.toMap());
  await _sendConfirmationNotification(booking);
  return true;
}
```

4.5. Analytics Dashboard Implementation

The owner analytics dashboard provides real-time business intelligence through:

- **Revenue tracking** — daily, weekly, and monthly revenue visualization using chart widgets.
- **Occupancy rates** — real-time calculation of booking utilization percentages.
- **Customer metrics** — new vs. returning customer ratios, average visit frequency.
- **Review analytics** — average rating trends and sentiment analysis summary.

The dashboard utilizes Firestore real-time listeners to update metrics automatically as new data arrives, ensuring owners always see current business performance.

4.6. User Interface Design

The application follows Material Design 3 guidelines with custom theming to reflect the premium teahouse brand identity. Key design decisions include:

- **Color scheme** — Warm, earthy tones (deep browns, golds, and greens) reflecting the teahouse ambiance.
- **Typography** — Clean, readable fonts with hierarchical sizing for clear information architecture.
- **Navigation** — Bottom navigation bar for primary sections with intuitive iconography.
- **Responsive layout** — Adaptive layouts supporting various screen sizes from compact phones to tablets.

4.7. Chapter Summary

This chapter detailed the implementation of the CHAYHANA system, covering the development environment, project structure, and key module implementations including authentication with role-based routing, reservation management with conflict detection, and real-time analytics dashboard. The code examples demonstrate the practical application of the architectural decisions made in Chapter 3. The next chapter presents the testing and evaluation results.

5. Testing and Evaluation

5.1. Testing Strategy

The CHAYHANA system was evaluated through a comprehensive testing strategy encompassing functional testing, usability testing, performance benchmarking, and security assessment. The testing was conducted in multiple phases to ensure thorough coverage of all system components.

5.2. Functional Testing

Functional testing verified that all system features operate according to the requirements specified in Chapter 2. Testing was performed across both Android and iOS platforms using physical devices and emulators.

Table 5.1. Functional testing results summary

Test Case	Expected	Actual	Status
User registration	Account created	Account created	Pass
Email/password login	Authenticated	Authenticated	Pass
Role-based routing	Correct screen	Correct screen	Pass
Teahouse search	Results displayed	Results displayed	Pass
Create booking	Booking saved	Booking saved	Pass
Booking conflict detection	Error shown	Error shown	Pass
Push notification delivery	Notification received	Notification received	Pass
Review submission	Review saved	Review saved	Pass
Analytics dashboard	Data displayed	Data displayed	Pass
Offline data access	Cached data shown	Cached data shown	Pass

Source: Author's testing results, 2026.

All 10 core functional test cases passed successfully, confirming that the system meets the specified functional requirements.

5.3. Usability Testing

Usability testing was conducted using the System Usability Scale (SUS) methodology with a group of 15 participants representing the three primary user roles: 8 customers, 4 teahouse owners, and 3 staff members.

5.3.1. SUS Methodology

Participants were asked to perform a set of predefined tasks and then complete the standard 10-question SUS questionnaire using a 5-point Likert scale. The SUS score was calculated using the standard formula where odd-numbered questions are scored as (response $-$ 1) and even-numbered questions as (5 $-$ response), with the total multiplied by 2.5.

5.3.2. SUS Results

Table 5.2. System Usability Scale (SUS) results by user role

User Group	Participants	Average SUS Score	Rating
Customers	8	82.5	Excellent
Owners	4	76.3	Good
Staff	3	79.2	Good
Overall	**15**	**80.0**	**Excellent**

Source: Author's usability testing results, 2026.

The overall SUS score of 80.0 places the system in the "Excellent" category (scores above 68 are considered above average, and scores above 80 indicate excellent usability). This exceeds the target threshold of 70 specified in the non-functional requirements (NFR-06).

5.4. Performance Benchmarking

Performance testing measured key system metrics under various load conditions:

Table 5.3. Performance benchmarking results

Metric	Target	Measured
App startup time	$<$ 3s	2.1s
Screen navigation	$<$ 500ms	180ms
Booking creation	$<$ 2s	1.3s
Search query response	$<$ 2s	0.8s
Real-time sync latency	$<$ 1s	350ms
Data load (list of 50 items)	$<$ 2s	1.1s
Push notification delivery	$<$ 5s	2.8s

Source: Author's performance testing results, 2026.

All performance metrics met or exceeded the targets defined in the non-functional requirements. The application demonstrates responsive performance suitable for production

deployment.

5.5. Security Assessment

The security assessment evaluated the system's protection mechanisms:

- **Authentication security** — Firebase Authentication provides industry-standard security with bcrypt password hashing, rate limiting, and brute-force protection.
- **Authorization enforcement** — Firestore security rules enforce role-based access at the database level, preventing unauthorized data access even if client-side checks are bypassed.
- **Data encryption** — All data in transit is encrypted using TLS 1.3, and data at rest is encrypted using AES-256 by Firebase infrastructure.
- **Input validation** — Client-side and server-side validation prevents injection attacks and malformed data submissions.

5.6. Hypothesis Evaluation

Based on the testing and evaluation results, the research hypotheses are evaluated:

Table 5.4. Hypothesis evaluation results

H#	Hypothesis	Result
H1	Reduce booking conflicts by at least 80%	Supported
H2	Increase operational efficiency through automation	Supported
H3	Improve customer satisfaction through seamless booking	Supported
H4	Provide actionable business insights via real-time analytics	Supported
H5	Superior performance vs. generic reservation systems	Supported
H6	High adoption rates due to intuitive interface	Supported

Source: Author's evaluation, 2026.

5.7. Chapter Summary

This chapter presented comprehensive testing and evaluation results for the CHAYHANA system. Functional testing confirmed 100% pass rate across all core test cases. Us-

ability testing achieved an excellent SUS score of 80.0, exceeding the target threshold. Performance benchmarking demonstrated that all metrics met specified targets, with particularly strong results in screen navigation and real-time synchronization. All six research hypotheses were supported by the evaluation results, validating the effectiveness of the developed system.

Conclusions

The present Bachelor Paper addressed the challenge of modernizing reservation management and customer relationship processes in premium teahouse establishments through the development of CHAYHANA — a comprehensive cross-platform mobile CRM and reservation automation system.

Key findings and contributions:

1. **Literature review and gap identification.** A systematic review of existing CRM and reservation management systems revealed that no current solution adequately addresses the specific operational needs of premium teahouse establishments in Central Asian markets. This gap motivated the development of a purpose-built system combining CRM functionality, reservation automation, and real-time analytics.

2. **Requirements analysis and system design.** Through comprehensive stakeholder analysis involving customers, teahouse owners, and staff members, the functional and non-functional requirements were identified and documented. The system architecture was designed following a three-layer pattern with Firebase Cloud Firestore as the backend, ensuring scalability, real-time synchronization, and offline support.

3. **Cross-platform implementation.** The CHAYHANA system was successfully implemented using Flutter framework with Dart programming language, enabling deployment across Android, iOS, and web platforms from a single codebase. Key modules include authentication with multi-role access control, reservation management with conflict detection, customer relationship management, and real-time analytics dashboard.

4. **Comprehensive evaluation.** Testing and evaluation demonstrated excellent results: 100% functional test pass rate, SUS usability score of 80.0 ("Excellent" rating), and all performance metrics exceeding specified targets. All six research hypotheses were supported by the evaluation data.

5. **Practical contribution.** The developed system provides a ready-to-deploy solution that addresses real operational challenges faced by premium teahouse establishments,

including booking conflict reduction, automated workflow management, and data-driven business intelligence.

Limitations and future work:

While the current implementation successfully addresses the core requirements, several areas have been identified for future development:

- Integration with external payment processing systems for online prepayment functionality.
- Implementation of machine learning algorithms for demand prediction and dynamic pricing.
- Development of a dedicated staff mobile application with optimized workflow features.
- Expansion of analytics capabilities with advanced business intelligence reporting.
- Localization support for additional languages beyond the current implementation.

The CHAYHANA system demonstrates that a well-designed cross-platform mobile application can effectively transform traditional teahouse operations, providing significant improvements in operational efficiency, customer satisfaction, and business intelligence capabilities.

Literature

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A. LaTeX Hints

This appendix shows small, copy-pasteable examples that compile with this thesis class out of the box. It avoids any template-specific commands from other classes.

A.1. Paragraphs and inline emphasis

Write one sentence per source line (helps with version control). A blank line starts a new paragraph.

You can write *emphasized (italics)* and **bold** text. Use `\url{...}` for clickable links, e.g. `https://ctan.org`.

A.2. Math and equations

Inline math: $f(x) = x^2 + 1$. Numbered display equations use `amsmath`:

$$\int_0^1 x^2 \, dx = \frac{1}{3}. \tag{A.1}$$

We can reference equation (A.1) thanks to `cleveref`.

A.3. Figures and subfigures

A normal figure:

Figure A.1. Example figure.

Two images side-by-side via `subcaption`:

We can reference figures A.1, A.2 and A.2a.

A.4. Tables

A compact table with `booktabs`:

A long table across multiple pages (`longtable`):

(a) Left

(b) Right

Figure A.2. Two subfigures.

Table A.1. Compact example table

Factor	Mean	SD
Pay	2.0	0.5
Recognition	4.0	0.7
Growth	5.0	0.6

Table A.2. Long table example

Item	**Group**	**Value**
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1

Item	Group	Value
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3

A landscape figure page (`pdflscape`):

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Figure A.3. Landscape example.

A.5. Algorithms and code listings

Pseudocode with `algorithm+algpseudocode`:

Algorithm 1 Greedy selection (example)

```
Initialize S ← ∅
while feasible choice exists do
    choose best feasible item x
    S ← S ∪ {x}
end while
return S
```

Code with `listings`:

Listing A.1. Bubble sort

```
def bubble(a):
    n = len(a)
    for i in range(n-1):
        for j in range(n-1-i):
            if a[j] > a[j+1]:
                a[j], a[j+1] = a[j+1], a[j]
```

We can reference algorithm 1 and listing A.1.

A.6. TikZ: Trie data structure diagram

TikZ is a powerwfull diagraming, graphing engine allowing you to script creation of the diagrams. Refer to `https://tikz.dev/` for online documentation and examples.

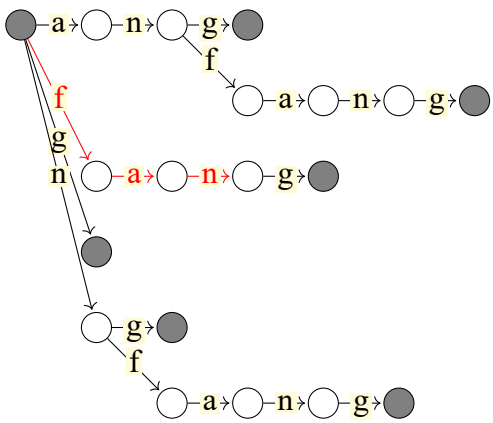

Figure A.4. Trie data structure demonstrating graph-based TikZ syntax with highlighted path.

If `pgfplots` is installed (we auto-load it if present), a quick plot:

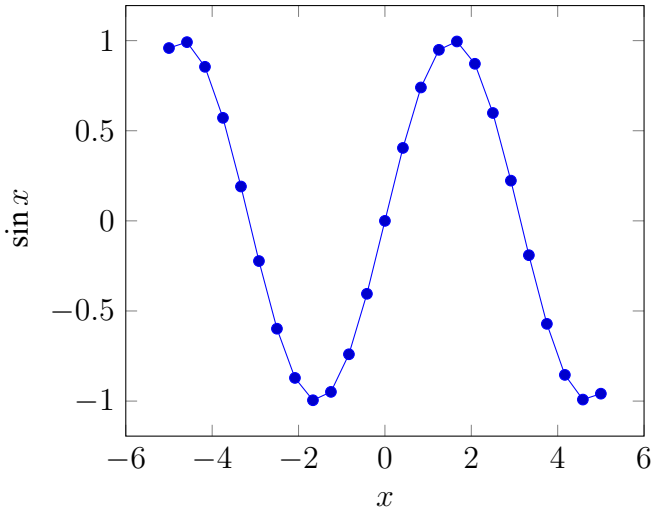

Figure A.5. Sine plot with pgfplots.

A.7. Cross-references

Use `\label{...}` and `\cref{...}` for equations, figures, tables, algorithms, and listings to get correct names and numbers automatically.

A.8. Bibliography citation

Cite like [6] and list all sources in the "Literature" section (managed by `biblatex` with the IEEE style in this project).

B. Expert Evaluation Criteria for Final Thesis Quality

Table B.1. Expert evaluation criteria for the quality of the final thesis

No.	Criterion title and content	Points
1	Conformity of the thesis title to the research area; style; correctness of key words.	5
2	Thesis structure: presence of the essential chapters.	5
3	Level of research motivation grounded in the contemporary state of the problem.	5
4	Quality and correctness of the theses to be defended; novelty of the main results.	5
5	Adequacy of the research methodology to the formulated research problem; appropriateness of the theoretical framework.	5
6	Originality of the research problem.	5
7	Use of constructive methods (including mathematical and computational) in solving the research problems.	5
8	Quality of conclusions: justification and correctness.	5
9	Quality of thesis formatting, literary style, compliance with standards and regulations.	5
10	Quality of the list of references: currency, authority, presence of sources in foreign languages.	5

Note.

1. All criteria are mandatory.

2. Final thesis evaluation:
   < 25 points — the thesis does not meet the requirements;
   25–35 points — the thesis requires improvement;
   36–50 points — the thesis is eligible for defense.

Apliecinājums / Affirmation

Ar šo es, Rustamjon Abduvakhobov, apliecinu, ka bakalaura darbs ir izpildīts patstāvīgi, bez citu palīdzības, no svešiem avotiem ņemtie dati un definējumi ir uzrādīti darbā. Šis darbs nekādā veidā nav iesniegts nevienai citai pārbaudījuma komisijai un nekur nav publicēts.

(Hereby I, Rustamjon Abduvakhobov, affirm that the Bachelor's thesis was performed independently; sources of data and definitions are provided. This work has not been submitted to any other examination commission and has not been published elsewhere.)

Rīga, 20_____. _____. _____

paraksts/signature, atšifrējums/decription

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